IP NAS Enclosure v6d

Print package for the print shop: 4 parts, PLA black, one plate, no supports, 0 slicer warnings



156.4 g	18.77 JOD	5 h 52 min	0
total filament (slicer)	at 0.12 JOD per gram	Bambu X1C estimate, one plate	slicer warnings

What is in this document

1. What to print (parts, files, weights). 2. Exact print settings. 3. Parts in print orientation. 4. Checks after printing. 5. How the parts go together. 6. Files in the package.

Companion document: **NAS_v6d_2D_Blueprints_A3.pdf** (5 dimensioned A3 sheets). Print files: folder **01_PRINT_FILES_STL** (4 STL files + a ready Bambu Studio / OrcaSlicer project **NAS_v6d_plate_Bambu_X1C_PLA.3mf**).

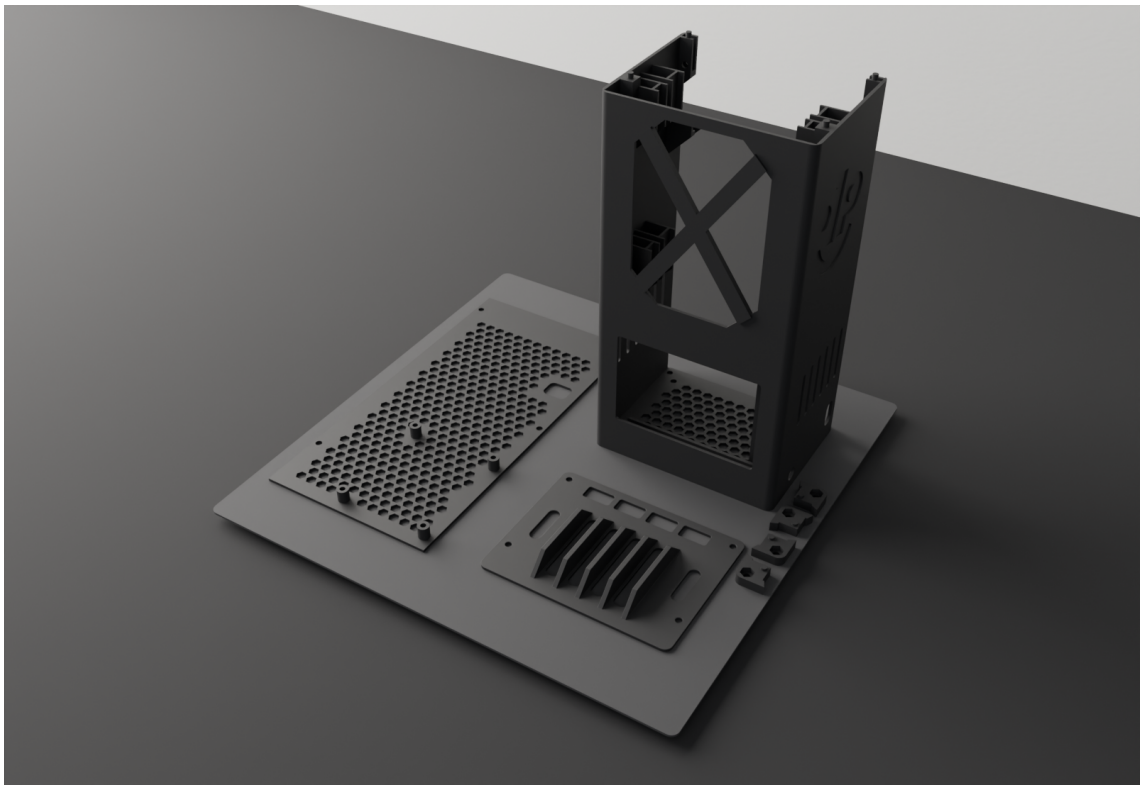
1. What to print

Four print files, all in **black PLA** (the legs file holds all 4 legs). The STL files are already in the correct print orientation and in millimetres: import them at 100 %, do not rotate, do not scale. Fastest route: open **NAS_v6d_plate_Bambu_X1C_PLA.3mf** in Bambu Studio or OrcaSlicer: all parts are already arranged and set up.

Part	File	Print orientation	Size on the bed (mm)	Filament	Time*
A Body	NAS_v6d_A_body.stl	Upright, open top up	100 x 79.2 x 197.5	110.0 g	3h 42m
B Back door	NAS_v6d_B_back_door.stl	Flat, outside face down, standoffs up	96.5 x 193.4 x 8.4	24.4 g	1h 32m
C Crown	NAS_v6d_C_crown.stl	Flat, plate down, fins up	100 x 78 x 11.6	18.0 g	26m
D Legs (x4)	NAS_v6d_D_legs_x4.stl	Flat, foot face down, pegs up	15.6 x 75.6 x 6.3	4.0 g	9m
TOTAL	4 files	one 256 x 256 plate		**156.4 g**	**5 h 52 min**

* Filament per part from PrusaSlicer 2.9.6 with the settings in section 2 (52.4 m of 1.75 mm). Cross-check in OrcaSlicer 2.4.2 with the stock Bambu Lab X1 Carbon profile: 153.1 g per part sum, 152.1 g and 5h 51m 32s for the whole plate (time column = whole plate, Bambu). Cost 156.4 g x 0.12 JOD = **18.77 JOD**. Optional 5 mm brim on the body: total 156.8 g.

Plate layout (all four files on one 256 x 256 bed)



Any arrangement works. The body is 197.5 mm tall, so the printer needs at least 200 mm of Z height. A bed smaller than 200 x 200 mm: print the back door (193.4 mm long) on its own plate.

2. Print settings

Use the shop's standard PLA profile and set these values. Everything not listed: printer default.

Setting	Value	Why
Material	**PLA, black**, 1.75 mm, any good brand	One colour only. No AMS colour changes, no purge waste.
Nozzle / layer height	0.4 mm / **0.20 mm** (first layer 0.20)	Wall thickness is designed as 1.6 mm = 4 lines.
Walls (perimeters)	**4**	Walls print solid; no infill inside 1.6 mm walls.
Top / bottom layers	**5 / 5**	Floor and plates are 1.6 mm = 8 layers.
Infill	**15 % gyroid**	Only bosses and thick corners get infill.
Supports	**OFF**	Designed support-free: overhangs are 45 deg or short bridges.
Brim	None. Optional 5 mm on the body only	Body is tall (197.5) on a 100 x 78 footprint. Adds 0.45 g.
Temperatures	Nozzle 210-220 C, bed 55-60 C	Standard PLA.
Cooling / bridges	Part fan 100 % from layer 3; bridge fan 100 %	Needed for the screen window bridge (see below).
Elephant-foot comp.	0.15 mm	Keeps the bottom edges and the leg holes true.
Scale / rotation	**100 %, as exported**	Fits are measured to 0.1 mm; do not scale.
Seam	Aligned, rear	Keeps the front face clean.

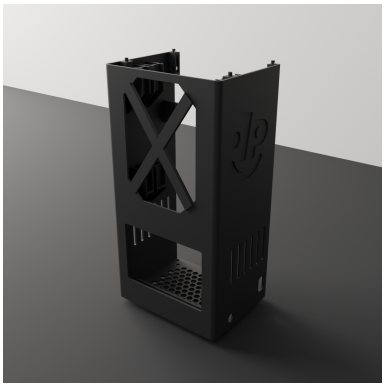
Slicer check, done before delivery: 0 warnings

Every file was sliced with supports OFF in two slicers. PrusaSlicer's stability check (floating anchors, long bridges, thin fragile parts) and Orca/Bambu's support check (floating regions, cantilevers) report **nothing** on any part or on the full plate. Overhangs are 45 deg or steeper; the screen window top (75.4 mm) is a normal bridge both slicers accept.

Fit check, done before delivery: 0 collisions, 5 of 5 assembly paths free. In the 3D model every printed part was tested against every other part and every bought part (Pi 5, both drives with their USB-SATA heads, screen, hub, screws, nuts), with all surfaces pulled in by 0.15 mm so only real overlaps count. Then the screen, each drive, the hub and the Pi door (with the USB and Ethernet plugs in) were slid along their real assembly path: nothing touches on the way in.

Part	PrusaSlicer 2.9.6	OrcaSlicer 2.4.2, Bambu X1C profile	Warnings
A Body	110.0 g, 4h 13m 31s	107.5 g, 3h 42m 2s	**none**
B Back door	24.4 g, 1h 22m 41s	23.7 g, 1h 32m 2s	**none**
C Crown	18.0 g, 29m 7s	18.1 g, 26m 27s	**none**
D Legs (x4)	4.0 g, 11m 52s	3.8 g, 9m 44s	**none**
Whole plate		152.1 g, 5h 51m 32s	**none**

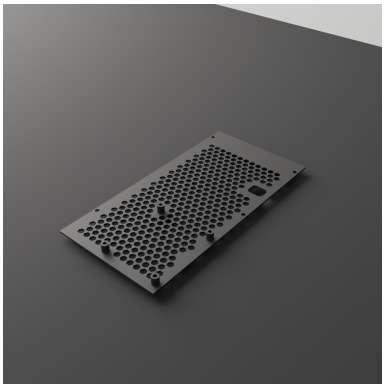
3. The four print files in print orientation



A Body

110.0 g, 4h 13m 31s

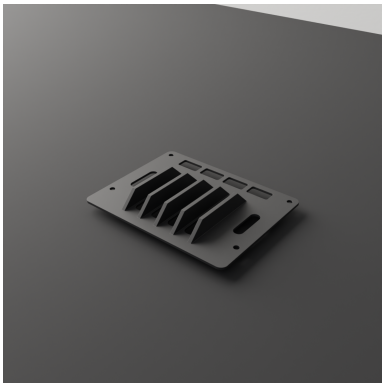
Upright, open top facing up, floor on the bed. Front: screen window and X-braced drive window. Right side: debossed logo, USB-C power window, 12 V hole. Inside: drive guide fins on both side walls with lanes, drive shelves at 91-93 mm, hub ledges at 151-153 mm, 4 drop-in M3 nut slots for the door, 4 pegs on the top rim for the crown, 4 holes in the floor for the legs.



B Back door

24.4 g, 1h 22m 41s

Flat, the smooth outside face on the bed, the four Raspberry Pi standoffs pointing up. Hex vents, Ethernet exit, 4 screw holes for M3.



C Crown

18.0 g, 29m 7s

Flat, plate on the bed, the five fins pointing up. 4 USB port windows at the back edge, exhaust slots, 4 peg holes.



D Legs (x4, one file)

4.0 g, 11m 52s

Four L-shaped corner legs, 5 mm high, foot face on the bed. On top: a hex pocket for an M3 nut and a 3 mm locating pin. Each leg is screwed on from inside the case (M3 x 6 through the floor into the nut), so the case stands 5 mm off the desk.

4. Checks after printing

Measure with a caliper before handing the parts over. These are the fits that matter.

#	Check	Where	Target	OK range
1	Body width / depth	outside, at the bottom	100.0 / 78.0	+-0.4
2	Body height to the rim	floor to top rim (not the pegs)	195.0	+-0.5
3	Drive lane 1	inside, gap between the front ribs	6.8	6.6 - 7.1
4	Drive lane 2	inside, gap between the back ribs	9.3	9.1 - 9.6
5	Screen slot	front wall to the bottom lip (centre)	6.3 deep	6.1 - 6.6
6	Crown fit	put the crown on the body	pegs 2.8 into holes 3.3	drops on by hand
7	Pi standoff pattern	door, centre to centre	49 x 58	+-0.3
8	Door holes	4 corners of the door	dia 3.5	M3 passes freely
9	Screen window top edge	front of the body	straight	sag < 0.5 mm
10	Vents clean	door hex vents, floor vents, side slots	no strings	clear by hand
11	Leg fit	M3 nut into the leg pocket, pin into the floor hole	nut 5.5 in 6.0, pin 3.0 in 3.5	drops in by hand
12	Nut slots	4 slots inside the body, at the back	M3 nut drops in	no force needed

Weights to expect on a scale

Part	Slicer weight	Accept
A Body	110.0 g	104 - 117 g
B Back door	24.4 g	23 - 26 g
C Crown	18.0 g	17 - 19 g
D Legs (4)	4.0 g	3.6 - 4.6 g
Total	**156.4 g**	149 - 166 g

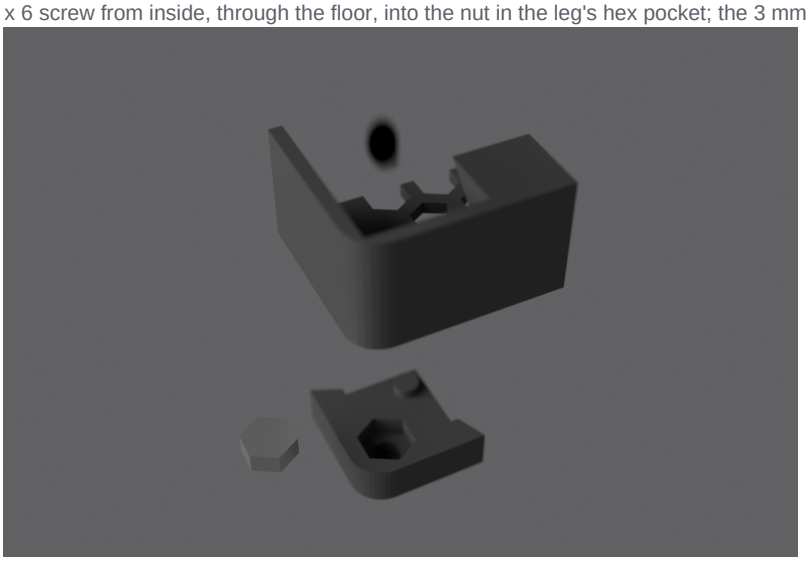
5. How the parts go together (for reference)



Not printed (bought)	Qty	Used for
M3 x 10 screw + M3 hex nut	4 + 4	back door to body (nuts slide into slots inside the body)
M2.5 x 12 nylon screw + M2.5 nut	4 + 4	Raspberry Pi 5 to the door standoffs
M3 x 6 low-head screw + M3 hex nut	4 + 4	legs: screw from inside the floor into the nut in each leg

Order: legs screwed on (M3 x 6 + nut each); screen slides into the front slot from the top; drives (with their USB-SATA heads) slide down the guide fins onto the shelves; M3 nuts dropped into their slots; Pi screwed to the door, cables plugged, door closed with 4 screws; hub drops onto its ledges; crown pressed onto the 4 pegs.

Leg fixing (exploded): M3 x 6 screw from inside, through the floor, into the nut in the leg's hex pocket; the 3 mm pin locates the leg.



6. Files in this package

Folder / file	What it is
01_PRINT_FILES_STL/NAS_v6d_A_body.stl	Body. Binary STL, mm, print orientation. 0 non-manifold edges.
01_PRINT_FILES_STL/NAS_v6d_B_back_door.stl	Back door. Binary STL, mm, print orientation. 0 non-manifold edges.
01_PRINT_FILES_STL/NAS_v6d_C_crown.stl	Crown. Binary STL, mm, print orientation. 0 non-manifold edges.
01_PRINT_FILES_STL/NAS_v6d_D_legs_x4.stl	The 4 legs in one file. Binary STL, mm, print orientation.
01_PRINT_FILES_STL/NAS_v6d_plate_Bambu_X1C_PLA.3mf	Ready project: all 4 files arranged on one plate, Bambu X1C 0.4 / 0.20 mm / PLA, sliced. Opens in Bambu Studio and OrcaSlicer.
02_DOCUMENTS/NAS_v6d_01_Print_Instructions_A4.pdf	This document.
02_DOCUMENTS/NAS_v6d_2D_Blueprints_A3.pdf	5 dimensioned A3 drawings: assembly, body, door, crown, plate layout.
03_BLENDER/NAS_v6d_assembly.blend	Full model (Blender 5.2), textures packed. Collections: 1 printed parts (assembled), 2 bought parts, 3 cables, 4 print-orientation copies, 5 studio.
04_3D_MODEL/NAS_v6d_assembly.glb	3D view of the assembly. Opens in Windows 3D Viewer or any web glTF viewer.
05_RENDERS/	All renders used here (front, back, inside, exploded, top, plate, part views).
case/ folder of the project repo	The Blender and Python scripts that generate the model, the STL files, the drawings and this PDF; the caliper measurements; the slicer and fit-check results.

Editing the design

The geometry is generated by script, so a change is one number and one command. Example: open **case/scripts/nas_scene_v6d.py** in the project repo, change a value (wall thickness **T**, lane clearance, window size), then run **blender -b -P scripts/nas_scene_v6d.py -- export**. It rebuilds the model, re-exports the four STL files, the views and the .blend. Direct edits in the .blend are also fine for small changes; export the edited part with File > Export > STL (scale 1000, selection only).

Measured parts this design is built on (my caliper numbers): drives 100 x 69.5 x 6.4 / 8.9; USB-SATA head 52.3 x 50 x 13.1 (47.8 outside the drive); 12 V plug 44; screen 94.6 x 61.8 x 12.3; USB hub 77 x 41.4 x 12.1. All of them: case/README.md in the project repo. Power test already done on the real hardware: both drives on the 12 V adapter wake together in 2.0 s, 5 of 5 runs, no USB resets.